

This is the right moment to rethink the analytic structure, because the original analyses were built around a symmetric design that your data do not fully support, and the three-question reframing in the document you have shared is actually a better fit for both your theoretical framing and your results. Let me build a revised strategy that preserves that logic while being more specific about your data and more honest about where the predicted dissociation holds.¹

1 Revised Analytic Strategy

1.1 Orienting Principle

The three-question structure maps cleanly onto your introduction's theoretical argument. RTs establish *who is faster* — they are a measure of efficiency, not morphological architecture. The N250 asks whether early form-based parsing is modulated by orthographic skill. The N400 asks whether later semantic integration is modulated by lexical-semantic knowledge. Individual differences are the diagnostic tool throughout, but their role shifts across questions: OS is foregrounded in Questions 1 and 2, LP is foregrounded in Question 3. This asymmetry should be explicit in how the models are structured and reported.

1.2 Handling Base Frequency and Family Size

This needs to be resolved before the models are specified, because the choice affects every analysis.

The problem: Base Frequency and Family Size are moderately correlated at the item level, they appear together in multiple models, and your results suggest they do different things — Base Frequency drives RT effects for both words and nonwords, while Family Size drives ERP effects. Carrying both as independent predictors throughout every model produces multicollinearity and dilutes the theoretical signal.

Recommended solution: keep them separate but assign them to questions

Rather than collapsing them into a composite, assign each variable its primary analytic home based on what your data and theory suggest it indexes:

- **Base Frequency** is your primary lexical familiarity index for **RT models** — it drives the behavioral effects and is interpretable as overall lexical availability. Family Size can be included as a covariate in RT models but should not be foregrounded.
- **Family Size** is your primary morphological structure index for **ERP models** — it drives the N250 and N400 effects and is interpretable as the richness of morphological family representation. Base Frequency can be included as a covariate in ERP models but should not be foregrounded.

¹Source document: `Pasted markdown.md` :contentReference[oaicite:0]index=0

This is not arbitrary — it is theoretically motivated by the distinction between token-level familiarity (Base Frequency) and type-level morphological structure (Family Size), and it is empirically supported by the pattern of your results. It also eliminates the high-order interactions that are currently making the models difficult to interpret.

If a reviewer pushes back, the composite PCA approach is your fallback — but try the separation first, because it preserves the theoretical distinction that your introduction is built around.

1.3 Core Question 1: Does Orthographic Sensitivity influence overall lexical decision efficiency?

Theoretical framing: RTs index processing speed and overall efficiency. The finding that OS predicts overall RT but does not modulate the structure of morphological effects is theoretically meaningful — it suggests that orthographic skill affects how fast the system operates without changing what it does.

Model structure:

Words:

$RT \sim \text{Base Frequency} + \text{Orthographic Sensitivity} + (1 + \text{Base Frequency} \mid \text{participant}) + (1 \mid \text{item})$

- Base Frequency as the primary lexical familiarity predictor
- Family Size included as a covariate but not interacted or foregrounded
- LP included once and reported as null — do not interact

Nonwords:

$RT \sim \text{Complexity} + \text{Base Frequency} + \text{Orthographic Sensitivity} + (1 + \text{Complexity} \mid \text{participant}) + (1 \mid \text{item})$

- Complexity as the primary morphological contrast
- Base Frequency as lexical familiarity covariate
- LP included once and reported as null

What to claim: OS predicts overall efficiency across both lexical categories.

1.4 Core Question 2: Does Orthographic Sensitivity modulate early morpho-orthographic parsing (N250)?

Theoretical framing: The N250 indexes early form-based processing at the morpho-orthographic level.

Model structure:

Words (N250):

$N250 \sim \text{Family Size} \times \text{Orthographic Sensitivity} + \text{Language Proficiency} + (1 \mid \text{participant:electrode}) + (1 \mid \text{item})$

Nonwords (N250):

$N250 \sim \text{Complexity} \times \text{Orthographic Sensitivity} + \text{Complexity} \times \text{Language Proficiency} + \text{Family Size} + (1 \mid \text{participant:electrode}) + (1 \mid \text{item})$

What to claim: Early morpho-orthographic parsing is modulated by orthographic skill.

1.5 Core Question 3: Does Language Proficiency shape semantic integration of morphological information (N400)?

Theoretical framing: The N400 indexes semantic integration.

Model structure:

Words (N400):

$N400 \sim \text{Family Size} \times \text{Language Proficiency} + \text{Orthographic Sensitivity} + (1 \mid \text{participant:electrode}) + (1 \mid \text{item})$

Nonwords (N400):

$N400 \sim \text{Complexity} \times \text{Family Size} \times \text{Language Proficiency} + \text{Orthographic Sensitivity} + (1 \mid \text{participant:electrode}) + (1 \mid \text{item})$

What to claim: Language Proficiency shapes how morphological family structure is recruited during semantic integration.

1.6 Summary of Model Structure

Question	Measure	Primary predictor	Primary moderator
Q1 Efficiency	RT Words	Base Frequency	OS
Q1 Efficiency	RT Nonwords	Complexity + Base Frequency	OS
Q2 Early parsing	N250 Words	Family Size	OS
Q2 Early parsing	N250 Nonwords	Complexity	OS, LP
Q3 Semantic integration	N400 Words	Family Size	LP
Q3 Semantic integration	N400 Nonwords	Complexity $\times$ Family Size	LP

1.7 What This Strategy Gains

The main advantage over the original analysis is that it replaces a symmetric design with an asymmetric design that reflects your theoretical predictions.

1.8 The Core Issue

Your rewritten introduction and the three-question analytic framework are well aligned, but the analyses were designed around an older symmetric structure.

1.9 What Needs to Change and Why

The models need to reflect asymmetry explicitly.

- RT models: remove morphological $\times$ individual difference interactions
- N250 models: OS primary moderator
- N400 models: LP primary moderator

1.10 What Does Not Need to Change

The introduction does not need substantial revision.

1.11 Recommended Order of Operations

1. Rerun RT models
2. Rerun N250 models
3. Rerun N400 models
4. Rewrite results section
5. Adjust aims if needed